Draw the block diagram of c-access memory fn. Why it is necessary & how does it improve the memory access time.

The block diagram of m-way low-order interleaved memory structure is shown in fig1 below.

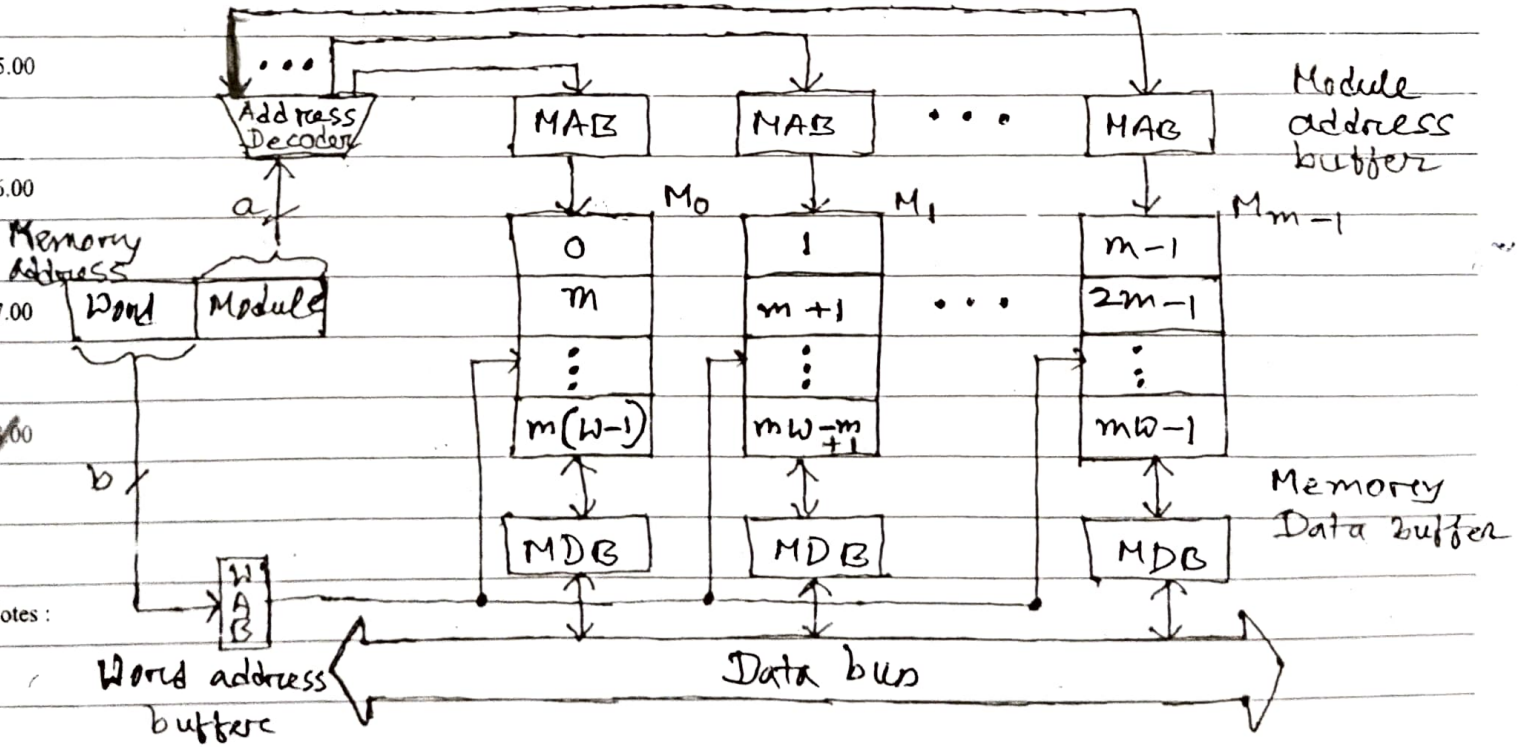


Fig 1: Low-order m-way interleaving (the c-access memory scheme)

✓ The concurrent access (C-access) of m-memory words in an overlapped manner is shown in fig-2.

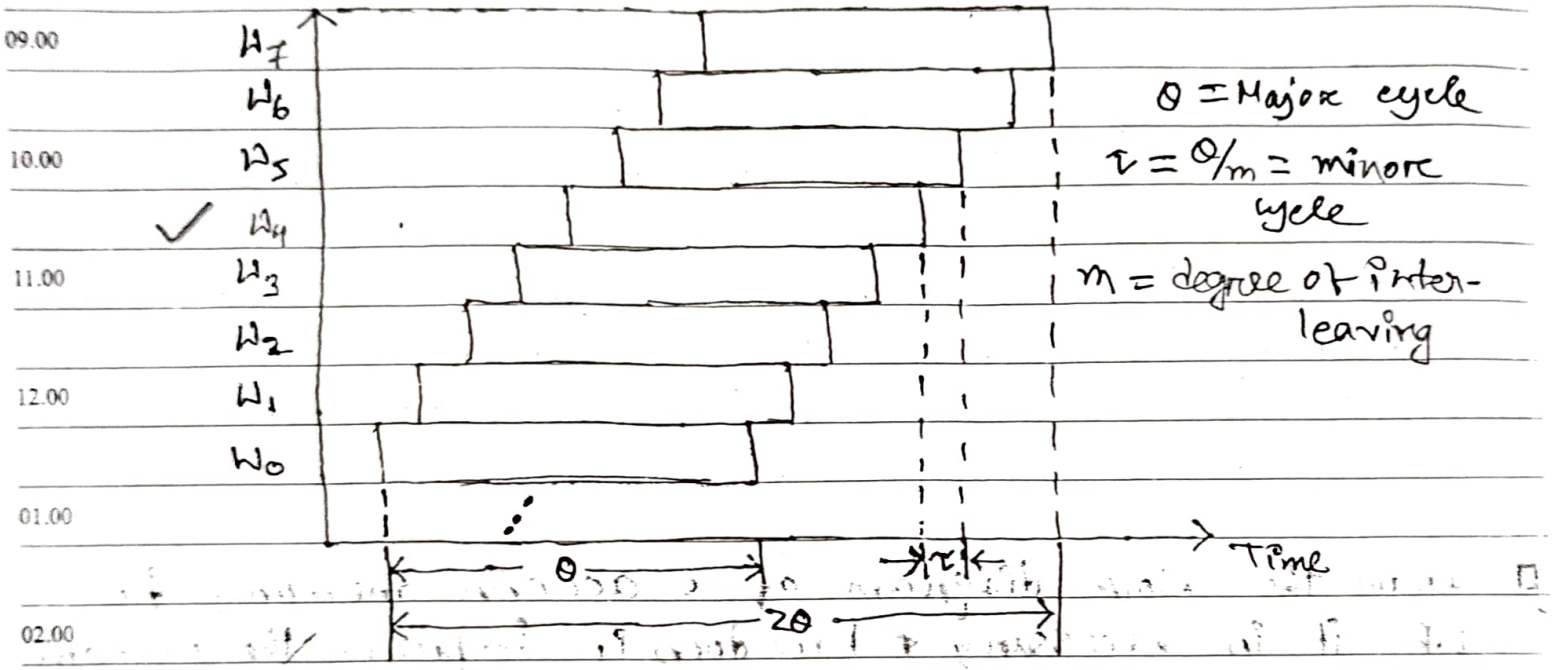


Fig 2: Pipelined access of eight consecutive words in a C-access memory.

✓ The low-order 'a' bits selects the modules, and the high-order 'b' bits select the word within each module, where $m = 2^a$ and $a+b = n$ is the address length.

✓ It is necessary to increase the throughput of the pipelined system.

✓ Because it access on vector. When the stride of the vector is 1, successive addresses are latched in the address buffer at the rate of one per cycle. It will take m minor cycles (= one major cycle) to fetch m-words. as shown in fig-2.

Notes

If the stride is 2, the successive accesses must be separated by two memory cycles in order to avoid access conflicts. This reduces the memory throughput by one-half.

If the stride is 3, there is no module conflict and the maximum throughput results.

In general, C-access will yield the maximum throughput of m-words per memory cycle if the stride is relatively prime to m, the number of interleaved memory modules.

□ Write the architecture and timing diagram of S-access memory organization.

In S-access (or simultaneous access) all memory modules are accessed simultaneously in a synchronized manner. The high-order (n-a) bits select the same word from each module.

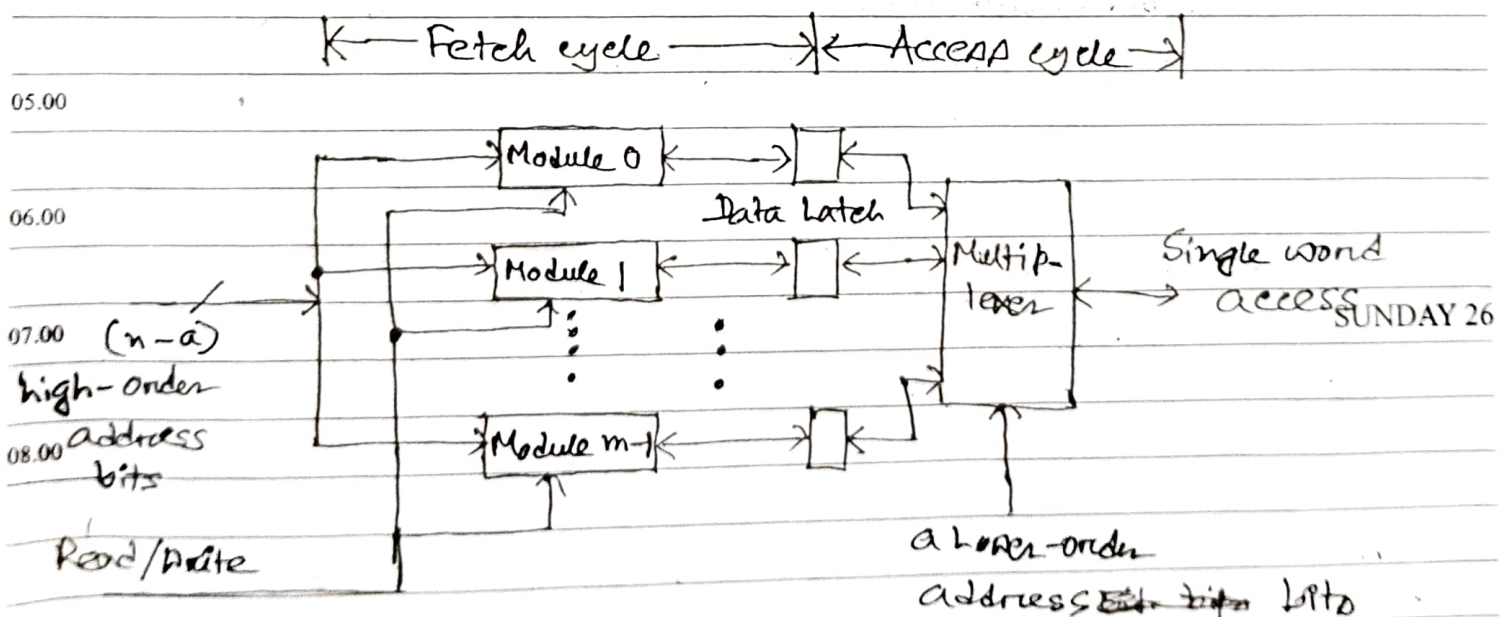
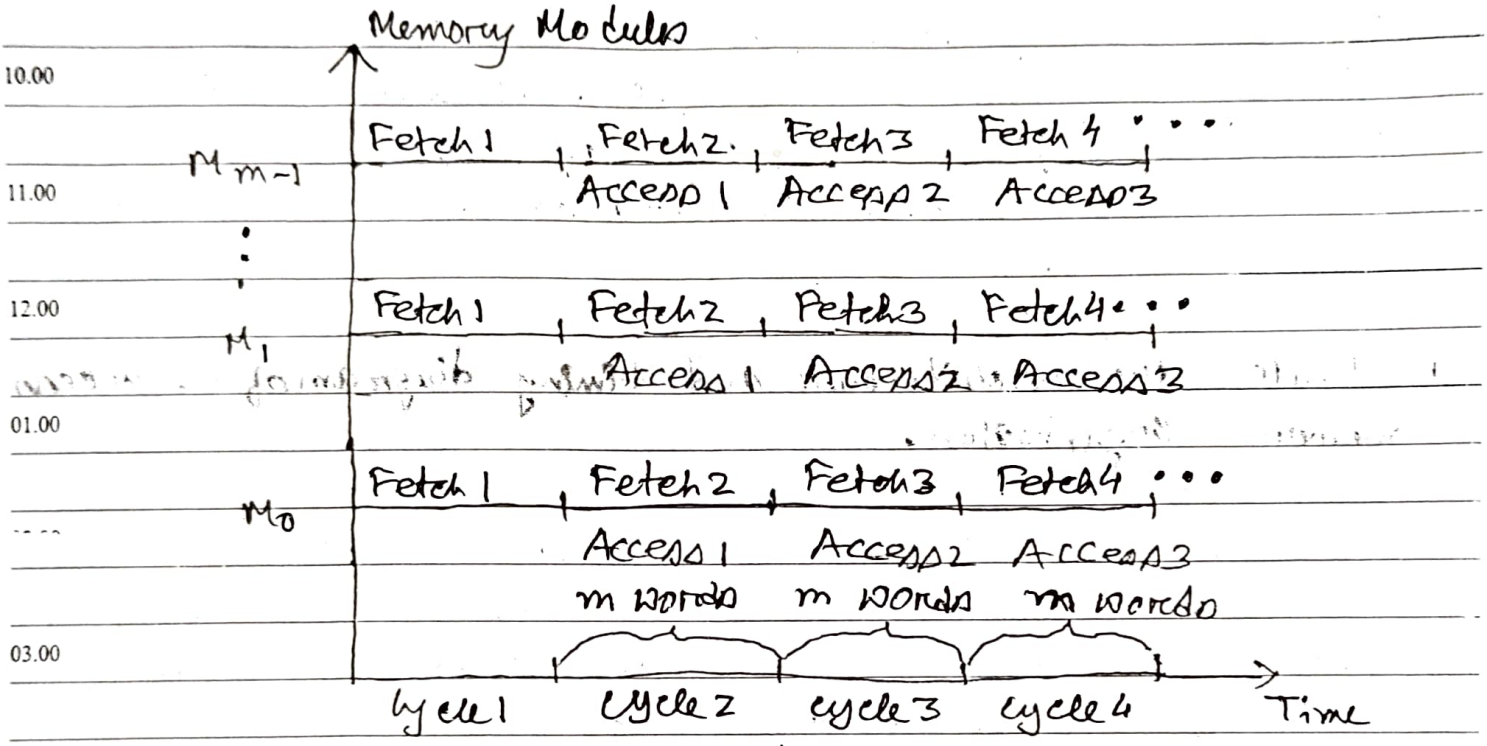


Fig 1:- S-access organization for an m-way interleaved memory.

- At the end of each memory cycle as shown in fig 2, $m = 2^a$ consecutive words are latched in the data buffers simultaneously.

09.00



10.00
11.00
12.00
01.00
03.00

04.00 Fig 2: Successive vector accesses using overlapped fetch and access cycles.

05.00

- The low-order a bits are used to multiplex the m words out, one per each minor cycle.

06.00

- If the minor cycle is chosen to be $1/m$ of the major memory cycle, then it takes two memory cycles to access m consecutive words.

07.00

- ~~However if~~ If the access phase is overlapped with the fetch phase of the current access, m words will take only one memory cycle.

08.00

Notes :

- If the stride is greater than 1, then the throughput decreases, roughly proportionally to the stride.